An Internship Report

Of

Core Java Virtual Internship

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE AI-ML

By

Roll Number of Student:

25SCS1003002909

Name of Student: ANUSHAKUMARI

Batch: 2025-2029

CANDIDATE'S DECLARATION

I, ANUSHA KUMARI, do hereby declare that the internship report titled "Core Java Virtual Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ANUSHA KUMARI

Roll No.: 25SCS1003002909

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the AICTE – UpSkill Campus Virtual Internship Program team and UniConverge Technologies Pvt. Ltd. for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Core Java. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-
Student Name: ANUSHA KUMARI
Roll No.: 25SCS1003002909

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3. INTERNSHIP COMPLETION CERTIFICATE

INTERNSHIP

CERTIFICATE OF COMPLETION

awarded to

Anusha Kumari

for successfully completing the Internship Program from **12-07-2026** to **26-08-2026**.

During this internship, the candidate actively participated in hands-on training and real industry project work in the domain of **Core Java**

This program was conducted in collaboration with **All India Council for Technical Education (AICTE)** and **UniConverge Technologies & upSkill Campus.**

Certificate ID: USC949986TIA

Kaushlendra Singh Sisodia
Chief Mentor, The IoT Academy
Director, Uniconverge Technologies

Internship Offer Letter

C-56/11, Ground floor, Near IT Stellar Park, Sector-62, Noida, U.P. - 201309
Email: support@upskillcampus.com, Ph. No. 9354068856

Internship Offer Letter

Date: **12-07-2026**

Name: **Anusha Kumari**
Email: **rohitkr128060@gmail.com**
Location: **Remote**

Dear **Anusha Kumari**,

We are pleased to inform you that **Upskill Campus** alongwith its industry Partner **UniConverge Technologies Pvt. Ltd.** has agreed to provide you internship as **Core Java Intern**
Your appointment of service is based on the following terms and conditions.

1. The internship shall start from **12-07-2026**
2. **Internship:** The internship period will be 4 to 6 weeks, starting from the date of joining.
3. **Working Hours:** The internship runs from 10:00 AM to 6:00 PM.

On behalf of both the Company, we wish you every success in your position and trust that our relationship will be long and mutually rewarding.

Sincerely
For Upskill Campus and UniConverge Technologies Pvt. Ltd.

UNICONVERGE TECHNOLOGIES PVT.LTD NOIDA

Prabha Singh
HR Manager

4. PROJECT DESCRIPTION

1. Introduction

This report presents a detailed account of the Virtual Internship undertaken under the AICTE – UpSkill Campus Internship Program in partnership with industry partner UniConverge Technologies Pvt. Ltd., in the domain of “Core Java Development”. The internship was carried out from 12th July 2026 to 26th August 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida.

The primary aim of this internship was to build a strong practical foundation in Object-Oriented Programming (OOP), the Java 8 Stream API, and Core Java File I/O, and to apply these principles to develop an industry-standard, persistent console application: the Console-Based Expense Tracker Application.

The internship followed a structured, milestone-driven curriculum combining conceptual understanding with hands-on software development, weekly progress reviews, and project implementation. On successful completion of all requirements, the official Virtual Internship Certificate of Completion (Certificate ID: USC949986TIA) was awarded by UniConverge Technologies and UpSkill Campus under the patronage of AICTE.

2. Organization Profile

UpSkill Campus and UniConverge Technologies Pvt. Ltd. (UCT) work in collaboration with the All India Council for Technical Education (AICTE) to deliver industry-aligned technical internships under the National Internship Portal.

UniConverge Technologies is an established technology solutions provider specializing in enterprise software, IoT, and embedded engineering. Through this partnership, the Core Java track was formulated to impart practical software engineering practices, clean architecture design (3-tier architecture), and robust coding standards to engineering graduates.

3. Problem Statement

Although a large number of engineering students learn the theoretical foundations of Object-Oriented Programming and Java syntax during academic coursework, very few receive structured, hands-on exposure to building modular, multi-tier software systems without relying on heavy external database management systems.

Concepts such as domain encapsulation, functional Stream pipelines, defensive copying with unmodifiable collections, multi-file stream persistence, and crash-proof console input validation are often understood only theoretically. The problem addressed by this internship was to practically design, architect, and implement an enterprise-patterned Console-Based Expense Tracker that reliably manages expense transactions, categorizes expenditures, enforces monthly budget thresholds, and persists data across sessions using Core Java streams.

4. Project Objectives

- To understand and apply first principles of Object-Oriented Programming (OOP) and Encapsulation to model financial transaction entities (Expense, Category).
- To implement a custom delimited CSV serialization and deserialization engine with delimiter-escaping mechanisms.

- To build a complete Expense CRUD Lifecycle (Add, View All, Selective Field Editing, and ID-based Deletion).

- To implement multi-criteria filtering by category and by calendar month (java.time.YearMonth) using declarative Java 8 Streams.

- To build categorized expenditure aggregation pipelines utilizing Collectors.groupingBy() and Collectors.summingDouble().

- To engineer a Monthly Budget Management Subsystem that dynamically calculates remaining balances and triggers real-time visual warning alerts upon budget breach.

- To implement resilient Flat-File Persistence (expenses.csv, budget.txt) using Core Java BufferedReader and BufferedWriter streams.

- To design an interactive, menu-driven Console UI (ConsoleUI.java) featuring fail-safe input validation loops that catch format exceptions (NumberFormatException, DateTimeParseException) without terminating the program.

- To create a 1-click execution script (run.bat) for automated compilation and instant execution.

4.5 Scope of the Project

The scope of this project encompasses the complete Core Java development lifecycle—from domain entity modeling and repository-based file stream persistence, to business logic processing using the Java 8 Stream API, and culminating in an interactive console-based user interface.

The application is engineered completely in Java SE (Standard Edition) without external database engines or third-party dependencies, demonstrating pure Core Java capabilities. The project was version-controlled and published on GitHub.

4.6 Technologies and Tools Used

- Core Language: Java SE 17+ (Core Java)
- Object-Oriented Design: Encapsulation, Abstraction, Data Hiding, and Static Factory Methods
- Java Collections Framework: ArrayList, List, Map, Collections.unmodifiableList()
- Functional Programming: Java 8 Stream API, Lambda Expressions, Method References
- Date & Time API (java.time): LocalDate, YearMonth, DateTimeFormatter
- File I/O Persistence: FileReader, FileWriter, BufferedReader, BufferedWriter
- Exception Handling: try-catch-finally, NumberFormatException, DateTimeParseException
- Development Environment & Tooling: Visual Studio Code, Git, GitHub, Windows Batch (run.bat)

4.7 System Architecture

The Console-Based Expense Tracker follows a decoupled 3-Tier Modular Architecture ensuring high cohesion and separation of concerns:

- Domain Model Layer (com.tracker.model):
  -Expense.java: Represents financial transactions with private fields (id, amount, category, date, description), custom toCsv(), and fromCsv() parsers.
  -Category.java: Represents classification groups (Food, Transport, Bills, Shopping, etc.).
- Data Persistence Layer:
  ExpenseRepository.java: Manages dual-file I/O streams for expenses.csv (transactions) and budget.txt (budget configuration), featuring automated CSV header detection and fallback recovery.
- Business Logic Layer :
  ExpenseService.java: Executes business rules, stream-based category filtering, YearMonth temporal querying, dynamic ID sequence deduction, and monthly budget evaluation.
- Presentation UI Layer :
  ConsoleUI.java: Manages the interactive 7-option main menu, category selector, smart field editing, spending breakdown percentage formatting, and looped input readers.
- Application Bootstrap Layer:
  ExpenseTrackerApp.java: Main entry point initializing repositories, services, and bootstrapping the console interface.
- Automation Layer:
  run.bat: Automated Windows batch script for one-click compilation and execution.

4.8 Methodology

The internship was executed following a structured, milestone-driven methodology combining theoretical review, modular software design, and hands-on coding over a 4-6 week duration:

Week	Milestone/Module	Technical Description
Week 1	Java Fundamental& Environment Setup	Practiced Java variables, control loops, methods, OOP basics, VS Code setup, and Git repository initialization.
Week 2	Domain Modeling & Expense CRUD	Created Expense.java with CSV parsing, ExpenseRepository for file saving, and initial expense recording.
Week 3	Stream Processing & Budget Subsystem	Implemented Java 8 Stream pipelines,

		Collectors.groupingBy(), YearMonth filtering, and budget limit tracking.
Week 4	Console UI, Automation & Final Release	Built ConsoleUI.java, real-time budget alert banner, run.bat script, and completed project documentation.
Week 5-6	Edge-Case Testing & Final Documentation	Performed comprehensive system verification, code optimization, final reporting, and GitHub release packaging.

4.9 Expected Outcomes

->Building a Complete Java Project:
Learned how to design a real working Java application from scratch by organizing code into separate, clean classes using Object-Oriented Programming (OOP).

->Working with Data easily :
Learned how to filter, search, and calculate totals (like monthly spending and category-wise breakdowns) using modern Java Streams instead of writing long, complicated loops.

->Saving Data without a Database (File Handling):
Learned how to save and load expenses automatically using text and CSV files (expenses.csv), so data is never lost even after closing the program.

->Preventing Application Crashes (Error Handling):
Learned how to handle user mistakes—like typing letters instead of numbers or entering wrong dates—by showing friendly error messages instead of the program crashing.

->Creating an Easy-to-Use Console Menu:
Built a clean, interactive menu system that lets users add, view, edit, delete expenses, and set monthly budget limits with automatic warning alerts.

->Version Control using Git & GitHub:
Learned how to manage project code week-by-week and publish the final completed project on GitHub.

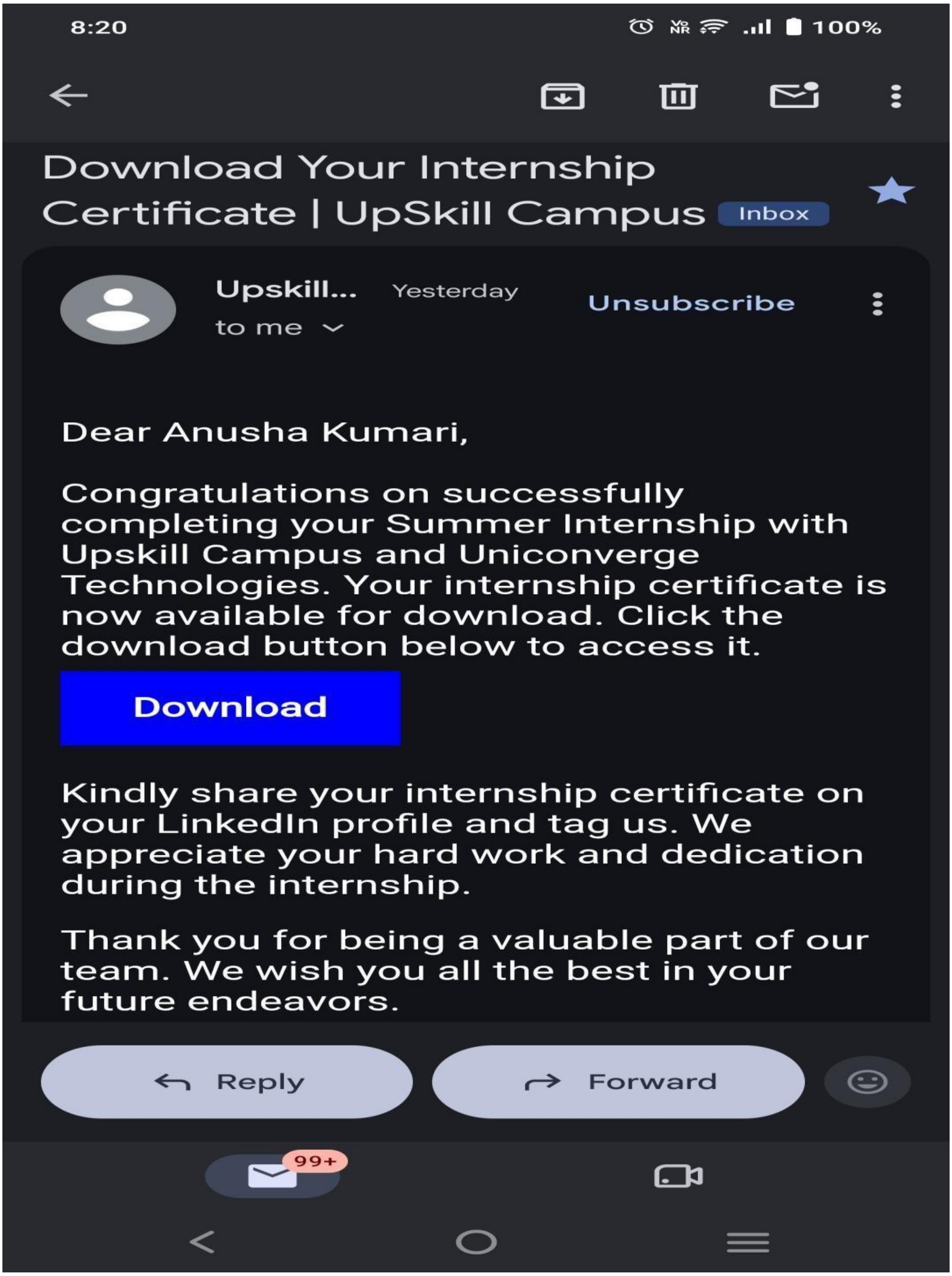
4.10 Certificates of Completion and Communication Proof

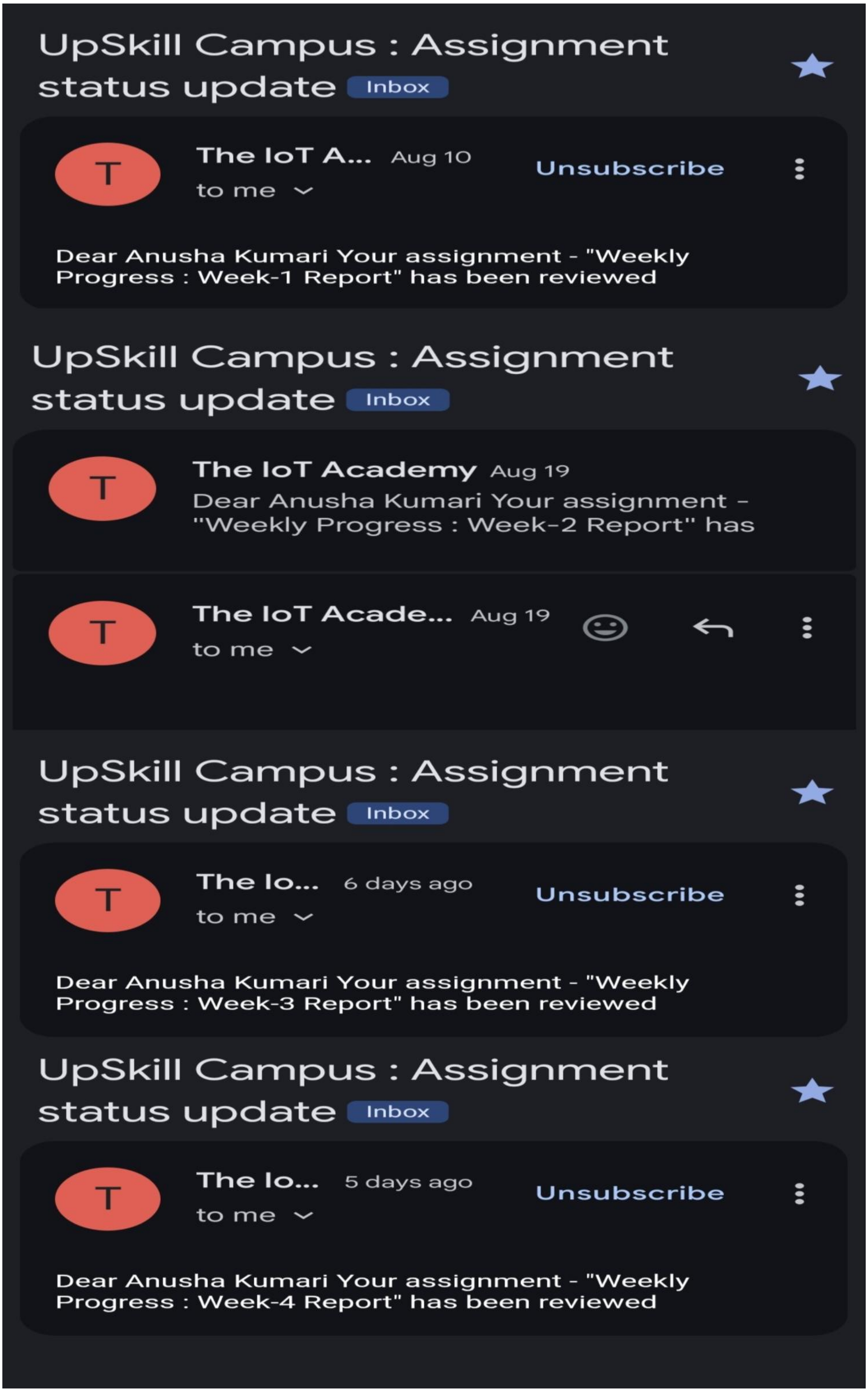
The official Internship Offer Letter and Certificate of Completion (Certificate ID:

USC949986TIA) issued by UniConverge Technologies Pvt. Ltd. and UpSkill Campus in collaboration with AICTE confirm the successful completion of the Core Java Virtual Internship.

The complete open-source project repository, source code, and commit history are publicly hosted on GitHub.

GitHub Link: https://github.com/anushasingh17/Expense-Tracker-Java

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